

Barangay Anao Web Portal

System Specification & Package Inventory

Generated Date: 2026-06-15	Barangay LGU: Anao, Ginatilan, Cebu, PH
Tech Stack: React 19, Vite 6, Tailwind v4	Framework Mode: Single Page Application (SPA)

1. Core Architecture Overview

The Barangay Anao Web Portal is an interactive, fully localized single-page website developed to serve the residents and administration of Barangay Anao. It features dual-language translations (English and Cebuano), smooth section-scrolling navigation with active-scrolling state hooks, an interactive hero panoramic carousel, a community news board, a sub-village (sitio) directory, local tourist gallery spotlight, and inquiry secretariat contact forms.

2. Integrated Tools & Code Quality Standards

- Unit Testing (Jasmine):** Configured using a TypeScript ESM executor (`tsx`) to run behavioral tests targeting the L10n translations parity, sitio metadata, and council members schema structures without requiring browser rendering overhead.
- Code Standards (ESLint):** Implemented flat configurations (`eslint.config.js`) combined with TypeScript and React Hook plugins to audit code quality, naming rules, and catch bugs during development.
- CI/CD Pipelines (GitHub Actions):** Configured pipeline (`.github/workflows/ci.yml`) checking dependency resolution, code lint compilation, Jasmine specifications, and production-ready static asset builds on every commits push or Pull Request.

3. Complete Packages Inventory

Package Name	Version	Type	Functional Description
react	^19.0.1	Production	Core component declaration and state management.
react-dom	^19.0.1	Production	Virtual DOM rendering to physical web browser views.
vite	^6.2.3	Production	Hot-module replacement dev server and bundler.
@tailwindcss/vite	^4.1.14	Production	Direct compiler plugin integrating Tailwind v4 stylesheet setup.
@vitejs/plugin-react	^5.0.4	Production	React compiler integration hooks for dev server.
lucide-react	^0.546.0	Production	SVG icons pack for location, contacts, and navigation.
motion	^12.23.24	Production	Interactive transition animations for sliders and dropdowns.
@google/genai	^2.4.0	Production	Google Gemini API developer SDK integration.
express	^4.21.2	Production	Node web application server framework for production deployment.
dotenv	^17.2.3	Production	Environment variables loader for secrets and config values.
typescript	~5.8.2	Development	Static type compiler and code editor assistant.
tsx	^4.21.0	Development	TypeScript execution shell for running Jasmine in ESM.
jasmine	^6.3.0	Development	Unit testing assertion suite and specs executor.
eslint	^9.39.4	Development	Javascript/TypeScript code standards reviewer.

Package Name	Version	Type	Functional Description
typescript-eslint	^8.61.0	Development	ESLint compiler extensions parser for TS syntax.
eslint-plugin-react	^7.37.5	Development	ESLint rule additions for React JSX component patterns.
eslint-plugin-react-hooks	^7.1.1	Development	ESLint rule check for React lifecycle hook callbacks.